

CURT GEN-AI Technical Questions

Answers

Question 1

Explain the difference between generative and discriminative models. Give one example of each.

Answer:

Discriminative models predict a label from input, like a spam filter deciding spam or not spam. Generative models learn to produce new data, like an LLM writing new text.

Question 2

What is the purpose of transformers in modern AI? How does it work briefly?

Answer:

transformers use attention to look at all words in a sequence at once and weigh relevance, instead of processing word-by-word like older models, making them faster to train and better at long-range context.

Question 3

Describe the difference between supervised, unsupervised, and reinforcement learning.

Answer:

supervised learning trains on labeled data input + correct answer, unsupervised learning finds patterns in unlabeled data on its own example clustering, and reinforcement learning has an agent learn by taking actions and getting rewards/penalties over time.

Question 4

What are tokens in the context of Large Language Models? Why are they important?

Answer:

tokens are the chunks like words, subwords, or characters LLMs break text into to process it, example "playing" might split into "play" + "ing." They matter because context limits, cost, and output length are all measured in tokens, not words.

Question 5

Explain the concept of attention in neural networks. How does it improve model performance?

Answer:

attention lets the model assign different importance weights to different words when processing each word, example in "the bank by the river," it helps connect "bank" to "river" to get the right meaning. This improves performance by capturing relationships between words regardless of how far apart they are in the text.

Question 6

What is prompt engineering, and why is it crucial for Gen AI applications? State the different types of prompts used.

Answer:

prompt engineering is crafting inputs to get better/more reliable outputs from an LLM, since the same model can perform very differently depending on how a question is phrased. Types: zero-shot which is no examples given, few-shot which is a few examples given, chain-of-thought which is asking the model to reason step by step, and role/system prompts setting context or persona.

Question 7

Explain how RAG (Retrieval-Augmented Generation) works.

Answer:

instead of relying only on what the model learned during training, RAG retrieves relevant documents/chunks from an external knowledge source usually a vector database at query time, then feeds those chunks into the prompt alongside the user's question. This lets the model answer using up-to-date or private information it was never trained on.

Question 8

What is embedding in AI, and why is it important for retrieval tasks?

Answer:

an embedding is a numerical vector representation of text that captures semantic meaning similar meanings end up close together in vector space. They're important for retrieval because you can compare a query's embedding to document embeddings using similarity search to find relevant content, instead of relying on exact keyword matches.

Question 9

How does temperature affect the output of a language generation model?

Answer:

temperature controls randomness in the model's output. Low temperature makes output more deterministic/focused, since the model tends to pick the highest-probability next token. High temperature increases randomness, producing more varied but less predictable output.

Question 10

What is chain-of-thought prompting, and why can it improve reasoning in LLMs?

Answer:

asking the model to explicitly "think step by step" before giving a final answer. This improves reasoning on multi-step problems like math or logic because it forces the model to generate intermediate reasoning steps instead of jumping straight to an answer, which reduces errors on complex tasks.

Question 11

What is the difference between RAG and fine-tuning? Give a use case for each.

Answer:

RAG injects external/current knowledge at query time without retraining the model good for frequently changing data example a live inventory or docs. Fine-tuning actually updates the model's weights on a specific dataset good for teaching a consistent style, format, or domain behavior that doesn't change often. Example: RAG for a support chatbot answering from live documentation; fine-tuning for a model that needs to always respond in a specific brand voice.

Question 12

Explain the possible improvements that can be applied to a Naive RAG (Retrieval-Augmented Generation) system, both pre-retrieval and post-retrieval.

Answer:

pre-retrieval improvements include better chunking strategy which is semantic chunking instead of fixed-size splits and query rewriting/expansion which is rephrasing the user's question to retrieve better matches. Post-retrieval improvements include re-ranking retrieved chunks by relevance before passing them to the LLM, and compressing/summarizing chunks so they fit the context window better.

Question 13

What factors should you consider when choosing an LLM for an application? Consider accuracy, latency, cost, context window, privacy, and reasoning ability.

Answer:

weigh accuracy/reasoning ability needed for the task, latency requirements which is real-time chat vs batch processing, cost per token at expected usage volume, context window size which is how much data needs to fit in one prompt, and privacy which is whether data can leave your infrastructure or needs to stay on-prem/self-hosted.

Question 14

How can you evaluate whether an LLM-generated answer is grounded in the provided documents?

Answer:

check whether claims in the answer can be traced back to specific retrieved passages which is citation matching, use a separate LLM as a judge to score faithfulness against the source documents, or compare the answer's claims against the source content directly to catch unsupported statements which is hallucination detection.

Question 15

How do you think computer vision could be integrated with Generative AI?

Answer:

multimodal models which is image + text can let an assistant answer questions about uploaded images or diagrams, generate images from text descriptions, or extract structured data from scanned documents/photos using OCR combined with understanding.

Question 16

Design an AI assistant for CURT.

The assistant should allow members from the Mechanical, Electrical, Software, and Business teams to ask questions about their team's documents and data.

It should:

- Search internal documents
- Answer questions using RAG
- Access structured data when necessary
- Cite its sources
- Handle different file types
- Prevent unauthorized access
- Be evaluated for accuracy

Tasks:

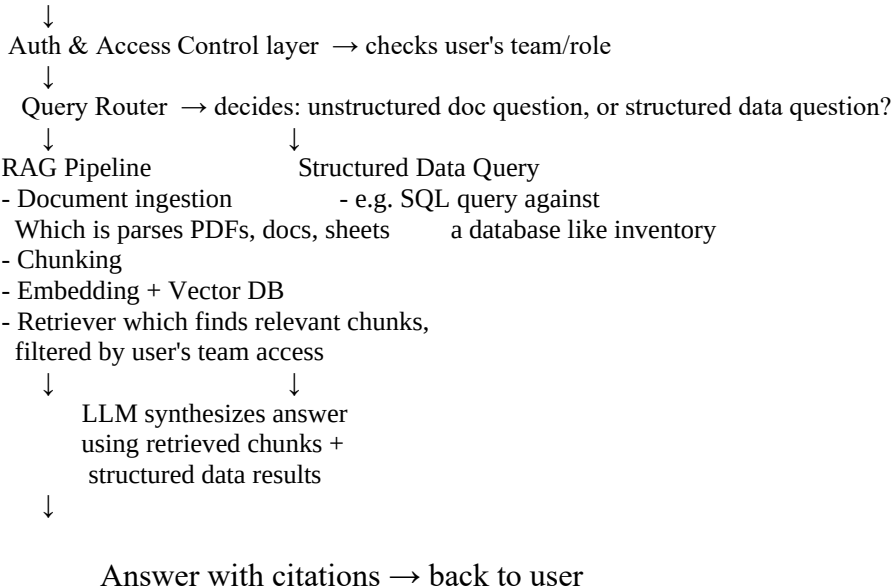
- a. Draw the system architecture and explain the purpose of each component.

- b. Identify the main challenges in the system related to: data preprocessing and handling different file formats; inconsistent, incomplete, or noisy data; chunking and token/context limits for large documents; retrieval of relevant information; accuracy of generated answers.
- c. Suggest at least two techniques or solutions to overcome these challenges.
- d. Explain how you would evaluate whether the system is working accurately and reliably.

Answer:

a.

Users like Mechanical / Electrical / Software / Business teams



b.

Data preprocessing / file formats: different parsers needed for PDFs, Word docs, spreadsheets, CAD files, images each requires different extraction logic, and some like CAD may not have clean text at all.

Inconsistent/noisy data: documents may have typos, outdated info, duplicate versions, or conflicting data across teams, which can mislead retrieval and generation.

Chunking/context limits: long documents must be split into chunks small enough to fit the model's context window, but splitting badly can cut information out of context example cutting a table in half.

Retrieval quality: the retriever might pull chunks that are topically similar but not actually relevant, or miss the right chunk if wording differs from the query.

Answer accuracy: the LLM can still hallucinate or misinterpret retrieved content even when given the right context.

c.

semantic chunking + re-ranking which is addresses chunking and retrieval quality, and metadata tagging/filtering by team and file type which is addresses access control and format handling,

plus citation-forcing prompts which addresses accuracy force the model to quote/reference the exact source it used.

d.

Sample real queries and manually check if answers match source documents (human review), track whether retrieved chunks actually contain the cited claim (automated groundedness check), and monitor user feedback/corrections over time as a live signal of accuracy.